1. The universal property of $S^{-1}A$. & localization.

Def IV. 1.1. S is a multiplicatively closed set with $1 \in S \subseteq A$ ring.

$S^{-1}A := A \times S / \sim$ where $(a,s) \sim (b,t)$ iff $(at-bs)u = 0$ for some $u \in S$.

Remark. 1). $\mathfrak{p}$ be the prime ideal of A. let $S = A - \mathfrak{p}$. then $A_{\mathfrak{p}} := S^{-1}A$ is a local ring.

In fact, the ideal $m = \{\frac{a}{s} : a \in \mathfrak{p}\}$ is the only maximal ideal in $A_{\mathfrak{p}}$. $A \to A_{\mathfrak{p}}$ called localization at $\mathfrak{p}$

2). let $f \in A$. $S = \{f^n\}_{n \geq 1}$. $A_f = S^{-1}A$.

3). $A = k[t_1, \dots, t_n]$, k a field. $\mathfrak{p}$ a prime ideal in A.

V is the variety defined by the ideal $\mathfrak{p}$. $= \bigcap_{f \in \mathfrak{p}} \operatorname{Ker} f(x)$

Def IV. 1.2. ① : M is an A-module. $S \subseteq A$. S... Then $S^{-1}M := M \times S / \sim$

where $(m,s) \sim (m',s')$ iff $\exists t \in S$. s.t. $t(s'm - sm') = 0$.

Then $S^{-1}M$ can be viewed as a $S^{-1}A$ - module

Similarly, we have $M_{\mathfrak{p}}$ and M_f.

② let $u: M \to N$ be an A-module homomorphism, then it gives $S^{-1}A$- module homomorphism: $S^{-1}u : S^{-1}M \to S^{-1}N$ $\quad m/s \longmapsto u(m)/s$.

Remark. A proper P is called a local property iff

A (or M) has $P \iff \forall$ prime ideals $A_{\mathfrak{p}}$ (or $M_{\mathfrak{p}}$) has P.

Prop. IV. 1.3. (the UMP of $S^{-1}A$): Let $g: A \to B$ ring homomorphism, s.t. $\forall s \in S$. $g(s)$ is a unit in S.

Then $\exists!$ h. s.t. $g = h \circ f$.

Pf:
$$\begin{array}{ccc} A & \xrightarrow{g} & B \\ {\scriptstyle f}\downarrow & \nearrow {\scriptstyle h} & \\ S^{-1}A & & \end{array}$$

i). Uniqueness: $h(a/1)$ & $h(1/s) \Rightarrow h(a/s) = g(a)g(s)^{-1}$.

ii) Existence: Let $h(a/s) = g(a)g(s)^{-1}$ ▨

Cora IV. 1.4. $g: A \to B$ satisfying: i) $s \in S \Rightarrow g(s)$ is a unit in B

ii) $g(a) = 0 \Rightarrow as = 0$ for some $s \in S$

iii) $\forall x \in B$. $\exists a \in A, s \in S$. s.t. $x = f(a)f(s)^{-1}$.

Then $\exists!$ h isomorphism. s.t. $g = h \circ f$.

pf: (skipped) ▨

Prop. IV. 1.5. If $M' \xrightarrow{f} M \xrightarrow{g} M''$ is exact at M,
then $S^{-1}M' \xrightarrow{S^{-1}f} S^{-1}M \xrightarrow{S^{-1}g} S^{-1}M''$ is exact at M''

pf: 1) $S^{-1}g \circ S^{-1}f = S^{-1}(g\circ f) = 0 \Rightarrow \mathrm{Im}(S^{-1}f) \subseteq \ker(S^{-1}g)$

2) $\forall\ m/s \in \ker(S^{-1}g) \Rightarrow g(m)/s = 0 \Rightarrow \exists\ t\in S$, s.t. $tg(m) = 0 = g(tm) \Rightarrow tm \in \mathrm{Im} f$

$\Rightarrow tm = f(m') \Rightarrow m/s = f(m')/st = (S^{-1}f)(m'/st) \in \mathrm{Im}(S^{-1}f)$ ▨

Com IV. 1.6. i) $S^{-1}(N+P) = S^{-1}(N) + S^{-1}(P)$

ii) $S^{-1}(N\cap P) = S^{-1}(N) \cap S^{-1}(P)$.

iii) $S^{-1}(M/N) \cong S^{-1}(M)/S^{-1}(N)$

iv. $S^{-1}A \otimes_A M \cong S^{-1}M$.

pf: Only check iii) Consider: $0 \to N \to M \to M/N \to 0$

Apply S^{-1}: $0 \to S^{-1}N \to S^{-1}M \to S^{-1}(M/N) \to 0$.

$\exists!\ \tilde{\Phi}$ isomorphism $S^{-1}(M)/S^{-1}(N)$ ▨

iv.) Define $\phi: S^{-1}A \times M \to S^{-1}M$ is A-bilinear

$(a/s, m) \mapsto am/s$

By UMP of $S^{-1}A \otimes M$, $\exists!\ f: S^{-1}A \otimes M \to S^{-1}M$. $a/s \otimes m \mapsto am/s$.

① surjective ✓

② injective: $\forall \sum_i (a_i/s_i)\otimes m_i \in S^{-1}A\otimes M$, let $s = \prod_i s_i$. $t_i = \prod_{j\neq i} s_j$

$\Rightarrow \sum_i (a_i/s_i)\otimes m_i = \sum_i \frac{a_i t_i}{s} \otimes m_i = \frac{1}{s} \otimes \sum_i a_i t_i m_i := \frac{1}{s} \otimes m$. $m \in M$.

Suppose that $m/s = 0 \Rightarrow \exists\ t\in S$. s.t. $tm = 0$. $\Rightarrow \frac{1}{s}\otimes m = \frac{t}{st}\otimes m = 0$ ▨

Corollary IV. 1.7. $S^{-1}A$ is a (flat) A-module

pf: Since the functor $S^{-1}(\cdot)$ keeps exactness and tensor product ▨

Prop. IV. 1.8. M, N: A-modules, $\exists!\ \phi$ isomorphism: $S^{-1}M \otimes_{S^{-1}A} S^{-1}N \to S^{-1}(M\otimes_A N)$ $\frac{m}{s}\otimes\frac{n}{t} \mapsto \frac{m\otimes n}{st}$.

In particular, $M_p \otimes_{A_p} N_p \cong (M\otimes_A N)_p$.

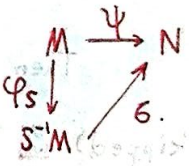
pf: Using 1.6.iv., we have $(S^{-1}M)\otimes_{S^{-1}A}(S^{-1}N) \cong (S^{-1}A\otimes M)\otimes(S^{-1}A\otimes N) \cong S^{-1}(M\otimes_A N)$ ▨

(where the last equation is common $(B\otimes M)\otimes(B\otimes N) \cong B\otimes(M\otimes N)$

$(b\otimes m)\otimes(b'\otimes n) \mapsto bb'\otimes(m\otimes n)$)

Prop. IV. 1.9 (UMP of $S^{-1}M$): M: R-module, given a R-homomorphism $\psi: M \to N$. $\exists!\ \sigma: S^{-1}M \to N$.

s.t. $\sigma\varphi_S = \psi$ where $\varphi_S: M \to S^{-1}M$ $m \mapsto m/1$



pf: i) Uniquess: $\forall m\in M, s\in S$, $\sigma(m/s) = \frac{1}{s}\sigma(m/1) = \frac{1}{s}\sigma\circ\varphi_S(m) = \frac{1}{s}\psi(m)$.

ii) Existence: ① set $\sigma(m/s) = \psi(m)/s$ for $\frac{m}{s} = \frac{n}{t}$ we have $u\in S$.

s.t. $utm = usn \Rightarrow \frac{\psi(m)}{s} = \frac{\psi(n)}{t} \Rightarrow \sigma$ is well-defined.

② σ is $S^{-1}R$-linear ▨

Prop IV. 1. 10. The functor $S^{-1}(\cdot)$ preserves direct limits.

Pf: $M = (M_i, \mu_{ij})$ is a direct system.

Denote that $S^{-1}M := (S^{-1}M_i, S^{-1}\mu_{ij})$ is a direct system.

We want to show that: $\varinjlim (S^{-1}M_i) \cong S^{-1}(\varinjlim M_i)$

~~Since $\forall i \in I$, $\exists \varphi_i : S^{-1}M_i \to S^{-1}(\varinjlim M_i)$ is a $S^{-1}R$ homomorphism~~

~~By the UMP of direct limits, $\exists !$ a $\varinjlim (S^{-1}M_i) \to S^{-1}(\varinjlim M_i)$ s.t. $S^{-1}\mu_i = a \circ S^{-1}\mu_i$ $\forall i \in I$~~

~~$\Rightarrow$ a is an isomorphism $\Leftrightarrow \varinjlim (S^{-1}M_i) \cong S^{-1}(\varinjlim M_i)$.~~

It suffices to show that $\bigoplus_i (S^{-1}M_i) \cong S^{-1}(\bigoplus_i M_i)$, $S^{-1}\mathrm{Coker}(\mu_i) \cong \mathrm{Coker}(S^{-1}\mu_i)$

then it's similar in the proof of $\otimes$ preserves direct limits ▨

Remark. S^{-1}, $\otimes$ are left-adjoint functor, so it preserves direct limits.

Prop. IV. 1. 11. The following are some local properties: M is an A-module. $\phi : M \to N$.

i) $M = 0 \Leftrightarrow M_p = 0$ $\forall$ prime ideals $p \Leftrightarrow M_m = 0$ $\forall$ maximal ideals m.

ii) ϕ injective $\Leftrightarrow \phi_p : M_p \to N_p$ injective $\Leftrightarrow \phi_m : M_m \to N_m$ injective

iii) M flat $\Leftrightarrow M_p$ flat $\Leftrightarrow M_m$ flat.

Pf: i) Only check ③ $\Rightarrow$ ①: If $M \neq 0$, choose $x \neq 0$. and let $a = \mathrm{Ann}(x) \neq (1)$.

$\exists$ maximal ideal $m \supseteq a$. then $M_m = 0$. and $x/1 \in M_m$

$\Rightarrow x/1 = 0 \Rightarrow \exists t \in A - m$. s.t. $tx = 0$. $t \in \mathrm{Ann}(x)$ ▨

ii). ① $\Rightarrow$ ② $\Rightarrow$ ~~③~~ $0 \to M \to N$ ~~inject~~ exact $\Rightarrow$ $0 \to M_p \to N_p$ exact $\Rightarrow$ $0 \to M_m \to N_m$ exact.

③ $\Rightarrow$ ①: $0 \to \mathrm{Ker}(\phi) \to M \to N$ exact $\Rightarrow 0 \to \mathrm{Ker}(\phi)_m \to M_m \to N_m$ exact

Also we have $\mathrm{Ker}(\phi)_m \cong \mathrm{Ker}(\phi_m) = 0 \overset{\text{(i)}}{\Rightarrow} \mathrm{Ker}\,\phi = 0 \Rightarrow \phi$ injective ▨

iii). ① $\Rightarrow$ ②: ~~The functor $S^{-1}(\cdot)$ preserves exactness~~ by extension of scalars, $f : A \to A_p$. $\Rightarrow M_p \cong A_p \otimes M$ is flat.

② $\Rightarrow$ ③: Obvious

③ $\Rightarrow$ ①: $N \to P$ injective $\Rightarrow N_m \to P_m$ injective

$\Rightarrow N_m \otimes M_m \to P_m \otimes M_m$ injective, M_m is flat.

$\Rightarrow (N \otimes M)_m \to (P \otimes M)_m$ injective, by IV. 1. 8.

$\Rightarrow N \otimes M \to P \otimes M$ injective by (ii). ▨

2. Prime spectrum of a ring and Zariski topology.

Def IV. 2.1. X be the sets of all prime ideals of A, $V(E)$ denote the set of all prime ideals containing E.

Then define the closed sets $V(E)$, then the resulting topology is called the Zariski topology., $X = \mathrm{Spec}(A)$

Remark 1. i) $V(0) = X \quad V(1) = \emptyset$

ii) $\bigcap_{i\in I} V(E_i) = V(\bigcup_{i\in I} E_i)$

iii) $\bigcup_{i=1}^{n} V(a_i) = V(\prod_{i=1}^{n} a_i) = V(\bigcap_{i=1}^{n} a_i)$ a_i are ideals

pf: Lemma: let a be the ideal generated by E. then $V(E) = V(a) = V(r(a))$.
(skipped)

i), ii) is obvious.

iii) just take $n=2$, $\mathfrak{p} \in V(a_1) \cup V(a_2)$ Assume $a_1 \not\subseteq \mathfrak{p} \Rightarrow a_1 \cap a_2 \subseteq \mathfrak{p} \Rightarrow \bigcup_{i=1}^{2} V(a_i) \subseteq V(\bigcap_{i=1}^{2} a_i)$

$\mathfrak{p} \in V(a_1 \cap a_2) \Rightarrow a_1 \cap a_2 \subseteq \mathfrak{p} \Rightarrow \mathfrak{p} \supseteq a_i$ (I. 1.8).

It's easy to check $\bigcup_{i=1}^{n} V(a_i) = V(\prod_{i=1}^{n} a_i)$.

2. $\forall f \in A$. let $X_f := \mathrm{Spec}(A) - V(f)$, then $\{X(f) \mid f \in A\}$ is a basis of the Zariski topology.

pf: Lemma: i) $X_f \cap X_g = X_{fg}$; ii) $X_f = \emptyset \Leftrightarrow f$ is nilpotent; iii) $X_f = X \Leftrightarrow f$ is a unit
(skipped)

$\forall\ Y = X \setminus V(E) = X \setminus V(\bigcup_{f\in E} f) = X \setminus \bigcap_{f\in E} V(f) = \bigcup_{f\in E} X_f$ ▨

~~Def IV.1~~ Prop. IV. 2.2. i) $\mathrm{Spec}(A)$ is compact.

pf:

ii) $\mathrm{Spec}(A)$ is a T_0-space.

iii) $\mathrm{Spec}(A)$ is connected $\Leftrightarrow e^2 = e \Rightarrow e = 0$ or 1.

iv). $\mathrm{Spec}(A)$ is irreducible $\Leftrightarrow$ the nilradical of A is a prime ideal.

v). $\{x\}$ is closed in $\mathrm{Spec}(A) \Leftrightarrow \mathfrak{p}_x$ is maximal

vi). X_f is homeomorphic (同胚) to $\mathrm{Spec}\, A_f$.

vii). $\phi: A \to B$ ring homomorphism, then $\phi^*: \underset{\| Y}{\mathrm{Spec}(B)} \longrightarrow \underset{\| X}{\mathrm{Spec}(A)}$. $\quad q \mapsto \phi^{-1}(q)$ is continuous

viii) $a \subseteq A$, $b \subseteq B$ be ideals. then $\phi^{*-1}(V(a)) = V(a^e)$. $\quad \overline{\phi^*(V(b))} = V(b^c)$.

In particular, $\begin{cases} \phi^*(Y) \text{ is dense in } A \Leftrightarrow \phi \text{ is injective.} \\ \phi \text{ surjective, then } \mathrm{Spec}(A) \cong \mathrm{Spec}(A/R). \text{ where } R = \mathrm{Nil}(A). \end{cases}$

pf:

Proof of Prop. IV.2.2:

(i). We prove a stronger thm: X_f is compact. let $X_f \subseteq \bigcup_{i\in I} X_{g_i}$.

Then $X_f = \bigcup_{i\in I} X_{fg_i} \Rightarrow V(f) = V(\underbrace{\{fg_i \mid i\in I\}}_{E})$ J is the ideal generated by E.

Thus $\forall \mathfrak{p} \supseteq J$ we have $f \in \mathfrak{p} \subseteq r(J) \Rightarrow \exists m\in\mathbb{N}_+$ s.t. $f^m \in J$.

$\Rightarrow \exists I'$ finite set s.t. $f^m = \sum_{i\in I'} f g_i \cdot h_i$, let J_0 be the ideal generated by $\{fg_i \mid i\in I'\}$

$\Rightarrow f^m \in J_0$.

Therefore, $\forall \mathfrak{p} \supseteq J_0$. we have $f^m \in \mathfrak{p} \Rightarrow f\in\mathfrak{p} \Rightarrow V(f) \supseteq V(\bigcup_{i\in I'}\{fg_i\})$ ▨

$\Rightarrow X_f \subseteq \bigcup_{i\in I'} X_{fg_i}$

(ii) (v.)Lemma: $\{x\}$ is closed in Spec(A) iff $\mathfrak{p}_x$ is maximal

pf: $\overline{\{x\}} = V(\mathfrak{p}_x) = \{x\}$ ⇐ $\mathfrak{p}_x$ maximal.

If $\mathfrak{p}_x$ isn't maximal. $\Rightarrow \exists$ maximal ideal $\mathfrak{q}$. s.t $\mathfrak{p}_x \subseteq \mathfrak{q} \Rightarrow V(\mathfrak{p}_x) \neq \{x\}$

Back to the proof, $y \in \overline{\{x\}} \Leftrightarrow \mathfrak{p}_x \subseteq \mathfrak{p}_y$

thus: $V(\mathfrak{p}_x) = V(\mathfrak{p}_y) \Rightarrow \mathfrak{p}_x = \mathfrak{p}_y$ ▨

(iii). If $e\in A$, $e^2=e$, $e\neq 0,1$. then $(1-e)^2 = 1-e$, $1-e \neq 0,1$.

We show that $X_e \cap X_{1-e} = \phi$. and $X_e \cup X_{1-e} = \mathrm{Spec}A$.

$e(1-e) = 0 \in \mathfrak{p} \Rightarrow e\notin\mathfrak{p}$, or $1-e\in\mathfrak{p}$, thus $X_e \cap X_{1-e} = \phi$. , $X_e \cup X_{1-e} = \mathrm{Spec}\,A$.

If $\mathrm{Spec}\,A = U \cup V$ U, V are open and closed $U\cap V = \phi$.

let $U = \bigcup_{i=1}^{m} X_{f_i}$. $V = \bigcup_{i=1}^{n} X_{g_i} \Rightarrow X_{f_i} \cap X_{g_j} = X_{f_i g_j} = \phi \Rightarrow f_i g_j$ is nilpotent.

J be the ideal generated by $\{f_i, g_j \mid 1\le i\le m,\ 1\le j\le n\}$. Then $J = A$.

$\Rightarrow \exists\ a\in\langle f_i \mid 1\le i\le m\rangle$. $b\in\langle g_j \mid 1\le j\le n\rangle$. s.t. $a+b=1$. $ab\in \mathrm{nil}(A)$.

$\Rightarrow a(1-a) = ab \in \mathrm{nil}(A) \Rightarrow \bar{a}^2 = \bar{a}$, $\bar{a}\in A/\mathrm{nil}(A) \Rightarrow \exists e\in A$. s.t. $e^2=e$ and $e-a\in\mathrm{nil}(A)$

$1-e \equiv b \pmod{\mathrm{nil}(A)}$ It's easy to check. $e\neq 0,1$. then $e^2=e$; $e\neq 0,1$ ▨

(iv.) X is irreducible $\Leftrightarrow \forall f,g\in A$. $X_{fg} = X_f \cap X_g \neq \phi$ ($X_f, X_g \neq \phi$)

$\Leftrightarrow \forall f,g \notin \mathrm{nil}(A)$, $fg \notin \mathrm{nil}(A) \Leftrightarrow \mathrm{nil}(A)$ is prime.

vi) $\phi : A \to A_f$ be the natural ring homomorphism.

induces ϕ^*: $\operatorname{Spec} A_f \to X_f$ $\quad q \longmapsto \phi^{-1}(q)$.

$f/1$ is a unit in $A_f \Rightarrow f \notin \phi^{-1}(q) \Rightarrow \phi^*(q) \in X_f$. (open)

$\Rightarrow \operatorname{Im} \phi^* \subseteq X_f$

$\forall\, \mathfrak{p} \in X_f$. we have $\phi^{-1}(\mathfrak{p}^e) = \mathfrak{p} \Rightarrow \phi^*$ is surjective.

$\forall\, \mathfrak{p}^e \neq \mathfrak{q}^e \in \operatorname{Spec} A_f$. $\phi^{-1}(\mathfrak{p}^e) = \mathfrak{p}$, $\phi^{-1}(\mathfrak{q}^e) = \mathfrak{q}$. $\Rightarrow \mathfrak{p} \neq \mathfrak{q} \Rightarrow \phi^*$ is injective.

By vii) We know that ϕ^* is continuous ▨

vii). $\forall\, V(a) \subseteq \operatorname{Spec}(A)$. $(\phi^*)^{-1}(V(a)) = \{\mathfrak{q} \in \operatorname{Spec} B \mid a \subseteq \phi^{-1}(\mathfrak{q})\}$. a is the ideal in A.

where $a \subseteq \phi^{-1}(\mathfrak{q}) \Leftrightarrow \phi(a) \subseteq \mathfrak{q}$.

Then $(\phi^*)^{-1}(V(a)) = V(\langle \phi(a) \rangle) = V(a^e)$ is closed. thus ϕ^* is continuous.

viii) is proved in the proof of vii). ▨

§3 Extended and contracted ideals in rings of fractions

Def IV.3.1. $f: A \to S^{-1}A$ be the natural homomorphism.

$a \subseteq A$, $b \subseteq S^{-1}A$ be ideals, then call $a^e := \langle f(a) \rangle$ be the extended ideal of a

$b^c := f^{-1}(b)$ be the contracted ideal of b

E be the sets of extended ideals, C be the set of contracted ideals.

Prop. IV.3.2. i). Every ideal in $S^{-1}A$ is an extended ideal. In fact, $b^{ce} = b$.

ii) $a^{ec} = \bigcup_{s \in S} (a : s)$, thus $a^e = (1)$ iff $a \cap S \neq \phi$.

iii). $a \in C \iff$ no element of S is a zero-divisor in A/a.

iv). $\mathfrak{p} \longleftrightarrow S^{-1}\mathfrak{p}$.

v). The functor S^{-1} commuts with finite sums, radicals, products, intersections.

pf:

i). $b \subseteq S^{-1}A$ ideal. $x/s \in b \Rightarrow x/1 \in b \Rightarrow x \in b^c \Rightarrow x/s \in b^{ce}$ thus $b \subseteq b^{ce}$.

$\Rightarrow b^{ce} = b$.

ii) $x \in a^{ec} = (S^{-1}a)^c \iff x/1 = a/s \iff \exists t \in S$. s.t $(xs - a)t = 0$. $\iff xst \in a \iff x \in \bigcup_{s \in S} (a : s)$

iii). $a \in C \iff a^{ec} \subseteq a \iff (sx \in a \Rightarrow x \in a) \iff$ no $s \in S$ is a zero-divisor in A/a

iv.). $S^{-1}A/S^{-1}\mathfrak{p} \cong \bar{S}^{-1}(A/\mathfrak{p})$.

v). (skipped)

Remark. 1. $a, b \subseteq A$ ideals, then $S^{-1}(a : b) = (S^{-1}a : S^{-1}b)$. b is finitely generated.

2. $R = \mathrm{nil}(A)$. then $S^{-1}R = \mathrm{nil}(S^{-1}A)$.

3. $\mathfrak{q} \subseteq \mathfrak{p}$ are prime ideals in $A \longleftrightarrow$ prime ideals of $A_{\mathfrak{p}}$.

Prop. IV.3.3. M is a finitely-generated A-module. Then $S^{-1}(\mathrm{Ann}\, M) = \mathrm{Ann}(S^{-1}M)$.

pf: M is generated by x. then $M \cong A/\mathrm{Ann}(M)$, $\Rightarrow S^{-1}M \cong (S^{-1}A)/(S^{-1}(\mathrm{Ann}\, M))$.

$\Rightarrow \mathrm{Ann}(S^{-1}M) = S^{-1}a = S^{-1}(\mathrm{Ann}\, M)$

$S^{-1}(\mathrm{Ann}(M+N)) = S^{-1}(\mathrm{Ann}(M) \cap \mathrm{Ann}(N)) = S^{-1}(\mathrm{Ann}(M)) \cap S^{-1}(\mathrm{Ann}(N)) =$ skipped. $\mathrm{Ann}(S^{-1}(M+N))$. ∎

Cora IV.3.4. $N, P \subseteq M$ submodules. P finitely generated, then $S^{-1}(N : P) = (S^{-1}N : S^{-1}P)$

pf: $(N : P) = \mathrm{Ann}\left(\frac{N+P}{N}\right)$. $\Rightarrow S^{-1}(N : P) = \mathrm{Ann}\left(S^{-1}\left(\frac{N+P}{N}\right)\right) = \mathrm{Ann}\left(S^{-1}(N+P) : S^{-1}N\right)$

$= (S^{-1}N : S^{-1}P)$

Prop. IV.3.5. $f: A \to B$ ring homomorphism. $\mathfrak{p} \subseteq A$ prime ideals. Then $\mathfrak{p}$ is the contraction of a prime idea of $B \iff \mathfrak{p}^{ec} = \mathfrak{p}$.

pf: $\Rightarrow$: let $\mathfrak{p} = \mathfrak{q}^c \Rightarrow$. $\mathfrak{p}^{ec} = \mathfrak{q}^{cec} = \mathfrak{q}^c = \mathfrak{p}$

$\Leftarrow$: $\mathfrak{p}^{ec} = \mathfrak{p}$ S be the image of $A \setminus \mathfrak{p}$ in B. then $\mathfrak{p}^e \cap S = \phi$.

$\Rightarrow S^{-1}\mathfrak{p}^e$ is a proper ideal contained in a maximal ideal $\mathfrak{m}$.

$\mathfrak{q} := \mathfrak{m} \cap B$. $\Rightarrow \mathfrak{q} \supseteq \mathfrak{p}^e$. $\Rightarrow \mathfrak{q}^c \supseteq \mathfrak{p}^{ec} = \mathfrak{p}$ and $\mathfrak{q} \cap S = \phi \Rightarrow \mathfrak{q}^c = \mathfrak{p}$. ∎

1. Topological groups and inverse limit.

Def V.1.1. G is an topological abelian group iff:

1). G is abelian;

2) G is a topological space;

3) $+: G\times G \to G$ and $-1: G \to G$ are continuous.
$(x,y) \mapsto x+y$ $\quad x \mapsto x^{-1}$

Def V.1.2. G is first countable. The set of all Cauchy equivalence classes is denoted by $\hat{G}$, the addition on $\hat{G}$ is $+: (x_n)+(y_n)=(x_n+y_n)$.

Then $\hat{G}$ is called the completion of G

Remark. 1). $\hat{G}$ is an Abelian group. $\phi: G\to\hat{G}$ $a\mapsto(a)$ is a homomorphism.
and $\ker\phi = \bigcap U$. U is the neighbourhood of 0.

2). G, H are topological Abelian groups. $f: G\to H$ continuous & homomorphism
then $\hat{f}: \hat{G}\to\hat{H}$: $(x_n)\mapsto(f(x_n))$ is the homomorphism.

Lemma V.1.3. G is Hausdorff $\Leftrightarrow$ $\{0\}$ is closed in G.

Lemma V.1.4. $H := \bigcap_{\substack{0\in U \\ U \text{ open}}} U$, then i) H is a subgroup.
ii) $H = \overline{\{0\}}$.
iii) G/H is a Hausdorff Abelian group.
iv) G Hausdorff $\Leftrightarrow H=\{0\}$.

pf: (skipped) ▨

Remark. 3). Suppose $G = G_0 \supseteq G_1 \supseteq \dots \supseteq G_n \supseteq \dots$ is a countable neighbourhood basis of 0. G_n are groups.
Then $G_n \subseteq G$ is closed and open subgroups.

Def V.1.5. Given a sequence of abelian groups $\dots \to A_{n+1} \xrightarrow{\theta_{n+1}} A_n \xrightarrow{\theta_n} \dots$
called this an inverse system.

$\varprojlim A_n :=$ the group of all coherent sequences (a_n) $\left(\theta_{n+1}(a_{n+1})=a_n,\ a_n\in A_n\right)$

Remark. 3) pro: now the inverse system is $\dots \to G/G_{n+1} \xrightarrow{\theta_{n+1}} G/G_n \xrightarrow{\theta_n} \dots$

where $\theta_{n+1}=\pi_{n+1}: G/G_{n+1} \to G/G_n \cong (G/G_{n+1})/(G_n/G_{n+1})$.

and we have $\hat{G} \cong \varprojlim G/G_n$. (*)

Example. $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$:

i) In algebra: $\mathbb{Z} \longleftarrow \mathbb{Z}/p\mathbb{Z} \leftarrow \mathbb{Z}/p^2\mathbb{Z} \leftarrow \dots \leftarrow \mathbb{Z}/p^n\mathbb{Z} \leftarrow \dots$ $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z} = \{(a_n)_{n\geq 1} \in \prod_{n\geq 1} \mathbb{Z}/p^n\mathbb{Z} \mid a_n \equiv a_{n-1} \bmod p^{n-1}\}$

ii) In topology: take the topology on $\mathbb{Z}/p^n\mathbb{Z}$ as the discrete topology

Then $\mathbb{Z}_p$ is compact, disconnected, metric, with none isolated points

Thus $\mathbb{Z}_p \cong C$ where C is the Cantor set. ▨

Prop. V.1.6. If $0 \to \{A_n\} \to \{B_n\} \to \{C_n\} \to 0$ is an exact sequence of inverse system,

then $0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n$ exact.

Moreover, $\{A_n\}$ is surjective system, then $0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0$. exact.

pf: $A = \prod A_n$. B, C similarly. and $d^A : A \to A$ $(a_n) \mapsto (a_n - \theta_{n+1}(a_{n+1}))$

then $\ker d^A = \varprojlim A_n$

Applying Snake-lemma on the following diagram:

$$\begin{array}{ccccccc} 0 & \to & A & \to & B & \to & C \to 0 \\ & & \downarrow d^A & & \downarrow d^B & & \downarrow d^C \\ 0 & \to & A & \to & B & \to & C \to 0 \end{array}$$

We have $0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to \operatorname{Coker} d^A \to \operatorname{Coker} d^B \to \operatorname{Coker} d^C \to 0$.

If $A_{n+1} \xrightarrow{\theta_{n+1}} A_n$ surjective, given $a_n \in A_n$ $\exists\, x_{n+1} \in A_{n+1}$ s.t. $x_n - \theta_{n+1}(x_{n+1}) = a_n$ (by induction) $\Rightarrow d^A$ = surjective ▨

Cora V.1.7. $0 \to G' \to G \xrightarrow{P} G'' \to 0$ exact. G have the topology defined by $\{G_n\}$.

ang give G', G'' the induced topology. $\{G' \cap G_n\}$, $\{pG_n\}$.

Then $0 \to \hat{G}' \to \hat{G} \to \hat{G}'' \to 0$ exact.

pf: notice that $0 \to \frac{G}{G' \cap G_n} \to \frac{G}{G_n} \to \frac{G''}{pG_n} \to 0$ exact. (It's easy to check).

Then by (*) in Remark 3) we have $0 \to \hat{G}' \to \hat{G} \to \hat{G}'' \to 0$ exact.

Remark. let $G' = G_n$., $G'' = G/G_n$. then G'' has the discrete topology (Since the basis are $G/G_n \supseteq G_1/G_n \supseteq \dots \supseteq \{0\}$. are finite. and by Remark 3). We have $\bar{g} \in G_i/G_n$ are open $\Rightarrow$ G'' has the discrete topology).

Cora V.1.8. $\hat{G}_n \trianglelefteq \hat{G}$ and we have $\hat{G}/\hat{G}_n \cong G/G_n$ by taking inverse limits we have $\hat{\hat{G}} \cong \hat{G}$.

pf of Prop. IV.2.2:

1)

Def V.1.7. Take $\mathfrak{t}$ as an ideal in A. $G = A$. $G_n = \mathfrak{t}^n$. then the topology defined on A is called the $\mathfrak{t}$-adic topology.

Remark. 1). $\mathfrak{t}$-adic topology is Hausdorff $\Leftrightarrow \bigcap \mathfrak{t}^n = (0)$.

2) $\phi: A \to \hat{A}$ is a cantinuous ring homomorphism, $\ker \phi = \bigcap \mathfrak{t}^n$.

Def V.1.10. M is an A-module. $G = M$. $G_n = \mathfrak{t}^n M$. This defines the $\mathfrak{t}$-topology on M.